# NEWT’S AND RAPH’S

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# Project Overview

This project aims to implement the Newton-Raphson Method for Optimization [1] polynomial function $f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots$, in matrix form:

$$P_{f(x)} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ \vdots \end{bmatrix}$$

It does so by multiplying it with a matrix constructed from the size of the polynomial in matrix form, the matrix for the derivative operation constructed being:

$$D_{\text{matrix}} = \begin{bmatrix} 0 & 1 & 0 & 0 & \ldots \\ & 0 & 2 & 0 & \ldots \\ & & 0 & 3 & \ldots \\ & & & 0 & \ddots \end{bmatrix}$$

Figure 1: The derivative operation for polynomials, represented as a derivative.

By multiplying them both, we get $f'(x)$ [2]:

$$P_{f'(x)} = D_{\text{matrix}} \times P_{f(x)}$$

Evaluating the polynomial at a given point is simple, by raising $x$ elementwise to the index of the polynomial in matrix form (the degree of each of the terms), then multiplying it to the coefficient defined in the polynomial matrix form.

$$f(x) = \sum_{i=0}^{n} x^{\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \\ n \end{bmatrix}} \circ \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix}$$

where $x^{\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \\ n \end{bmatrix}}$ is the elementwise operation of raising $x$ to every element from $1$ to $n$

# Objectives

- Develop a library that can calculate the Newton-Raphson method and other prerequisites operations (differentiation and polynomial evaluation).
- Provide a simple user interface that can expose the functionality of the library to a non-command line user.
- Check if the library correctly calculates the optimal values of a given function, up to tolerance.

## Showcase

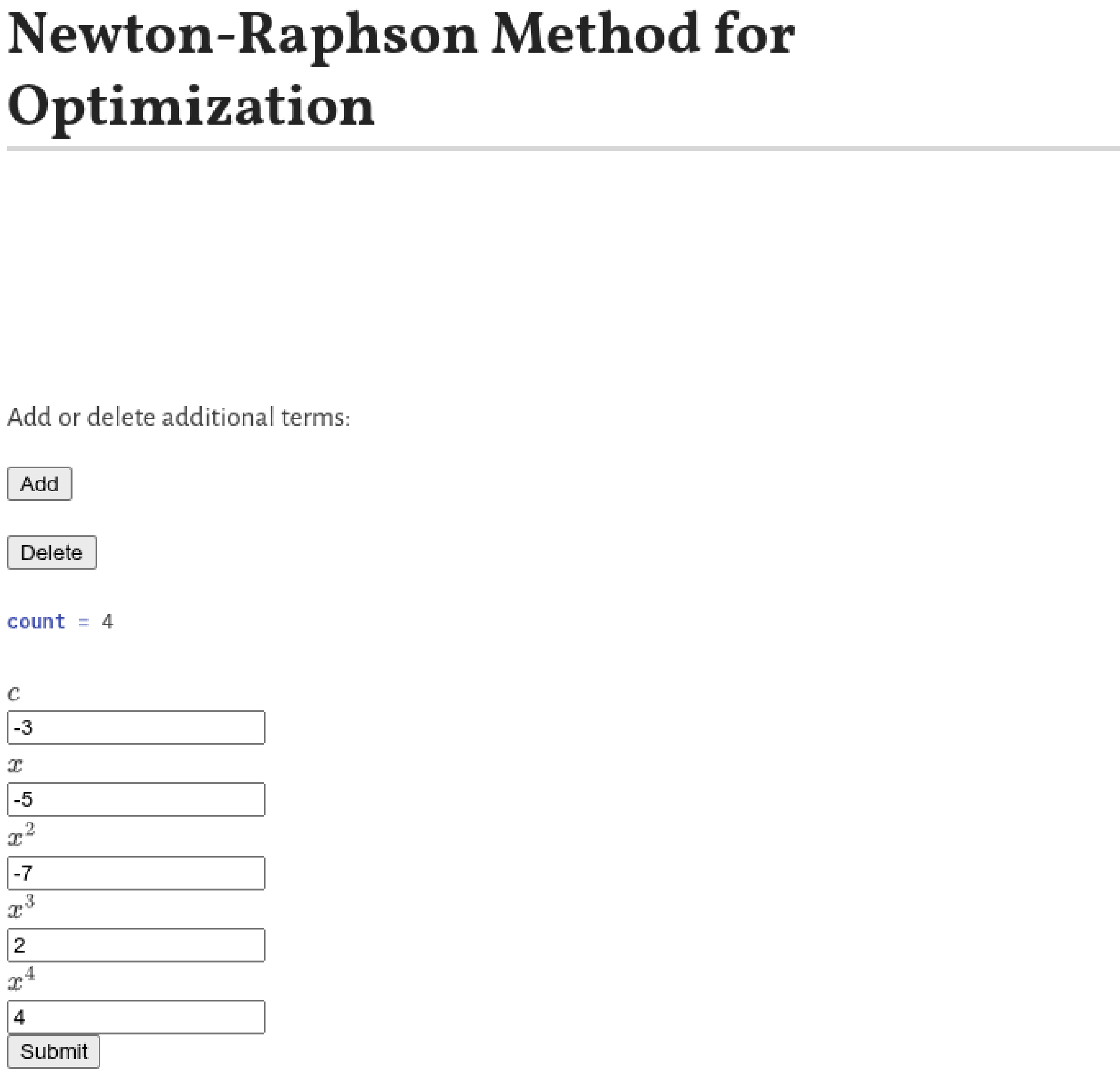



Figure 2: The title bar, plus the inputs for each of the terms of the polynomial.

## Polynomial

$$f(x) = -3.0 - 5.0x - 7.0x^2 + 2.0x^3 + 4.0x^4$$

## Derivative

$$f'(x) = -5.0 - 14.0x + 6.0x^2 + 16.0x^3$$

## Second Derivative

$$f''(x) = -14.0 + 12.0x + 48.0x^2$$

Figure 3: Show the input function, typeset with LaTeX.

0

Initial Guess

0.05

Tolerance

Figure 4: Enter initial guess and tolerance.

## Newton-Raphson Method for Optimization

Polynomial: $f(x) = -3.0 - 5.0x - 7.0x^2 + 2.0x^3 + 4.0x^4$

Inital Guess: $x_1 = 0.0$

Tolerance: $\epsilon = 0.05$

Optimal value: $-0.35714285714285715$

Optimal solution: $-2.133173677634319$

Figure 5: Display results of Newton-Raphson method for optimization

☑ Display table of optimization

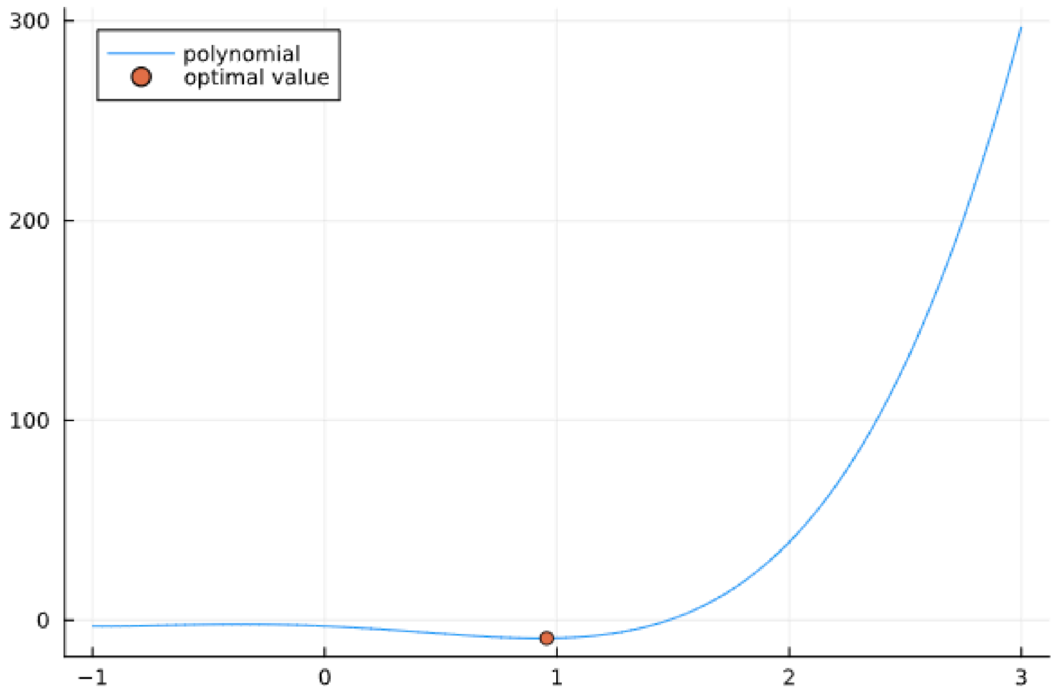
☑ Plot optimal values

-1

Min

3

Max




4×2 DataFrame

| Row | x_i<br>Float64 | ϵ<br>Bool |
|---|---|---|
| 1 | 2.03304 | false |
| 2 | 1.43059 | false |
| 3 | 1.09435 | false |
| 4 | 0.955976 | true |

Figure 6: Display graph and table.

## Assets

- GitHub Link

  To run, it needs the Julia programming language, and the Pluto Julia library.

- UI only (code will **not** run)

## Recommendations

- Extend the solver to be able to calculate derivatives for trigonometric and exponential functions.
- Implement an os-native user interface.

# Citations

[1] "Newton's method in optimization." Accessed: Dec. 15, 2025. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Newton%27s_method_in_optimization&oldid=1311976880

[2] "Derivatives via matrix multiplication." [Online]. Available: https://youtu.be/SSPbgRQCqUo